Pallerla Kartheek Reddy

Aspiring Gen AI Engineer

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Education

Godavari Global University — Rajahmundry
B.Tech in Computer Science & Engineering (AI & ML), CGPA: 9.02 — *2024 – 2028*

Sri Saraswati Junior College — Ongole
Intermediate (MPC), 94.2% — *2022 – 2024*

ZP High School — Budawada
SSC, 86.4% — *2020 – 2021*

Projects

HR Policy RAG Chatbot | *Python, LangChain, FAISS, HuggingFace, Groq, Streamlit* Live Demo | GitHub

- Built a Retrieval-Augmented Generation (RAG) chatbot that answers employee HR queries (leave, reimbursements, benefits) using only company policy PDFs.
- Implemented document chunking with `RecursiveCharacterTextSplitter`, embeddings with `all-MiniLM-L6-v2`, and semantic retrieval using a FAISS vector index.
- Added a guardrail prompt that classifies queries as in-scope or out-of-scope, so unrelated questions are declined and answers stay grounded in retrieved context.
- Used the Groq-hosted `gpt-oss-20b` LLM for answer generation and deployed the app on Streamlit Community Cloud.
- Scored **75.29** on the leaderboard of the Gen AI Project 2: Intelligent RAG challenge.

AI Voice Assistant | *Python, FastAPI, Gemini API, SpeechRecognition, gTTS, JavaScript* Live Demo | GitHub

- Built a full-stack voice assistant that takes speech input from the browser, generates responses with the Google Gemini API, and plays them back as speech.
- Developed a FastAPI backend with a speech-to-text, LLM, and text-to-speech pipeline using SpeechRecognition and gTTS.
- Created a browser interface (HTML, CSS, JavaScript) to record audio and interact with the assistant.
- Managed API keys securely using environment variables.

Agentic RAG E-commerce Support Agent | *Python, LangChain, LangGraph, FastAPI, Vector DB In Progress*

- Building an agentic RAG support agent that checks order status, calculates refunds, and processes returns for an e-commerce use case.
- Grounding responses in three policy documents (returns, shipping, product information) using vector embeddings and a vector database.
- Designing a LangGraph workflow where the LLM decides between policy retrieval and tool calls on a mock order database (order, refund, returns, policy lookup).
- Exposing the agent through a FastAPI backend.

Technical Skills

Languages: Python, JavaScript, HTML/CSS, SQL
GenAI / LLM: RAG, Agentic RAG, LangChain, LangGraph, Tool Calling, Prompt Engineering, Embeddings, Guardrails, Google Gemini API, Groq API
Vector DB & NLP: FAISS, HuggingFace Sentence-Transformers, Speech-to-Text, Text-to-Speech
Frameworks & Tools: FastAPI, Streamlit, React, REST APIs, MySQL, Git/GitHub

Training & Certifications

NxtWave CCBP 4.0 Academy — Pursuing
4-year program in software development and Gen AI

- Completed courses: Introduction to Databases, Python, Responsive Web Design, Static Web Design.